CEBU INSTITUTE OF TECHNOLOGY
UNIVERSITY

IT342-G1
System Integration and Architecture

System Design Document (SDD)

Project Title: BoduBodu - Bodyweight Exercise

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EXECUTIVE SUMMARY

1.1 Project Overview

BoduBodu – Bodyweight Exercise is a fitness website and mobile application designed to help users perform structured bodyweight workouts without the need for gym equipment. The platform provides guided exercise programs, workout tracking, and progress monitoring to support users in achieving their fitness goals anytime and anywhere.

The system consists of a Spring Boot backend API, a React-based web application, and a Kotlin-based Android mobile app. All components are integrated through RESTful APIs to ensure a consistent and seamless experience across platforms.

BoduBodu focuses on simplicity, accessibility, and user motivation, making bodyweight training structured, trackable, and beginner-friendly.

1.2 Objectives

1. Develop a fully functional bodyweight exercise platform with user authentication and workout management
2. Implement a three-tier architecture using Spring Boot (backend), React (web), and Android (mobile)
3. Create RESTful APIs to handle workout programs, exercise data, and user progress tracking
4. Design a responsive user interface that works consistently across web and mobile platforms
5. Enable users to track completed workouts and monitor fitness progress
6. Enable users to create and customize their own workout routines based on personal fitness goals.
7. Deploy all system components to production-ready environments

1.3 Scope

Included Features:

- User registration and authentication (email and password with JWT security)
- User profile management
- Exercise library containing bodyweight exercises with descriptions and instructions
- Categorized workout programs (Beginner, Intermediate, Advanced)
- Custom workout creation (users can create and modify their own workout routines)
- Workout session tracking (start workout, mark exercises as completed, finish session)
- Workout history and progress tracking
- Admin panel for managing exercises and workout programs
- Responsive web application
- Native Android mobile application
- MySQL relational database for structured data storage

Excluded Features:

- Integration with wearable fitness devices (e.g., smartwatches)
- AI-generated personalized workout plans
- Real-time video coaching or live streaming
- Email notification system
- Social media login integration
- Push notifications
- Paid subscription or payment gateway integration
- Advanced fitness analytics and performance prediction

1.0 INTRODUCTION

1.1 Purpose

This document serves as the comprehensive System Design Document (SDD) for the BoduBodu – Bodyweight Exercise system. It defines the system architecture, functional and non-functional requirements, database design, API specifications, and implementation plan.

The purpose of this document is to provide a clear technical blueprint that guides the development, integration, and deployment of the system across web and mobile platforms. It ensures that all components—including backend services, frontend applications, and the database are properly structured and consistently implemented.

2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

2.1 Project Overview

Project Name: BoduBodu - Bodyweight Exercise
Domain: Health/Fitness
Primary Users:

1. General Users (Fitness Users)
2. Administrators

Problem Statement: Many individuals want to stay active but do not have access to gym equipment or prefer equipment-free training. There is a need for a simple system that provides structured bodyweight workouts and progress tracking.
Solution: BoduBodu provides a cross-platform fitness system that offers structured bodyweight workout programs, customizable routines, and progress tracking through web and mobile applications.

2.2 Core User Journeys

Journey 1: New User Completing a First Workout

1. A new user accesses the web or mobile application.
2. The user registers using email and password, then logs into the system.
3. Available predefined workout programs are displayed.
4. A workout program is selected and its exercises are reviewed.
5. The workout session starts.
6. Exercises are completed and the session is finished.
7. The completed workout is saved in the user's workout history.

Journey 2: Creating a Custom Workout

1. The user logs into the system.
2. The “Create Custom Workout” section is accessed.
3. Exercises are selected from the exercise library.
4. Repetitions, sets, and rest intervals are configured.
5. The custom workout is saved.
6. The workout is started and completed.
7. The session is recorded in workout history.

Journey 3: Tracking Progress

1. The user navigates to the workout history section.
2. Previously completed workouts are displayed.
3. Details such as completion date and duration are reviewed.
4. The user may begin another predefined or custom workout.

Journey 3: Administrator Managing Exercises

1. The administrator logs in with authorized credentials.
2. The admin dashboard is accessed.
3. Exercises are added, edited, or removed.
4. Changes are reflected immediately in the system.

2.3 Feature List (MoSCoW)

MUST HAVE

1. User authentication (register, login, logout)
2. Exercise library (view exercises with details)
3. Predefined workout programs (Beginner, Intermediate, Advanced)
4. Custom workout creation and editing
5. Workout session tracking (start, complete, record duration)
6. Workout history tracking
7. Admin panel for exercise and workout management
8. MySQL database integration
9. Secure role-based access control (User / Admin)

SHOULD HAVE

1. Exercise filtering by difficulty level
2. Basic progress summary (e.g., total workouts completed)
3. Form validation with clear error messages
4. Responsive design across different screen sizes
5. Password recovery via email

COULD HAVE

1. Favorite workouts feature
2. Workout streak tracking
3. Simple progress statistics (e.g., total workout duration)
4. Minor UI enhancements and animations

WON'T HAVE

1. Email notification system
2. Social media login integration
3. Wearable device integration
4. AI-generated workout generation
5. Payment or subscription system
6. Push notifications
7. Admin user management

2.4 Detailed Feature Specifications

Feature: User Authentication

- **Screens:** Registration, Login
- **Fields:** Email, Password, Confirm Password
- **Validation:** Email format, password strength, uniqueness,
- **API Endpoints:** POST /auth/register, POST /auth/login, POST /auth/logout
- **Security:** JWT tokens, password hashing with bcrypt

Feature: Exercise Library

- **Screens:** Exercise Listing, Exercise Detail
- **Display:** Exercise name, description, difficulty level, target muscle group
- **API Endpoints:** GET /exercises, GET /exercises/{id}
- **Admin Functions:** POST /exercises, PUT /exercises/{id}, DELETE /exercises/{id}

Feature: Predefined Workout Programs

- **Screens:** Workout Listing, Workout Detail
- **Functions:** View workout details, Start workout session
- **Display**: Workout name, difficulty level, list of exercises
- **API Endpoints:** GET /workouts, GET /workouts/{id}

Feature: Custom Workout Creation

- **Screens:** Create Custom Workout, Edit Custom Workout
- **Field:** Workout name, selected exercises, repetitions, sets, rest interval
- **Functions:** Add exercises, Remove exercises, Update repetitions and sets, Save custom workout
- **API Endpoints:** POST /custom-workouts, GET /custom-workouts, PUT /custom-workouts/{id}, DELETE /custom-workouts/{id}

Feature: Workout Session Tracking

- **Screens:** Workout Session, Workout Summary
- **Data Collected:** Workout ID, User ID, completion date, duration
- **Process:** Start workout, Complete workout, Save session record
- **API Endpoints**: POST /sessions, GET /sessions/user/{id}

Feature: Exercise Library

- **Screens:** Exercise Listing, Exercise Detail
- **Display:** Exercise name, description, difficulty level, target muscle group (multiple), exercise video
- **API Endpoints:** GET /exercises, GET /exercises/{id},
- **Admin Functions:** POST /exercises, PUT /exercises/{id}, DELETE /exercises/{id}

Feature: Admin Panel

- **Screens:** Admin Dashboard, Exercise Management, Workout Management
- **Functions:** Add/edit/delete exercises, Add/edit/delete predefined workouts
- **Access Control:** Admin role required
- **API Endpoints:** Admin-prefixed endpoints with role validation

2.5 Acceptance Criteria

AC-1: Successful User Registration

- ➢ **Given** a new user is on the registration page
- ➢ **When** valid email and password are entered

- **And** the email is not already registered
- **And** the registration form is submitted
- **Then** a new account shall be created
- **And** the user shall be able to log into the system

AC-2: Successful User Login

- **Given** a registered user is on the login page
- **When** correct email and password are entered
- **And** the login form is submitted
- **Then** the system shall authenticate the user
- **And** generate a valid JWT token
- **And** redirect the user to the dashboard or homepage

AC-3: Viewing Exercise Library

- **Given** a logged-in user accesses the exercise section
- **When** the exercise list is loaded
- **Then** all available exercises shall be displayed
- **And** each exercise shall show its name and difficulty level

AC-4: Creating a Custom Workout

- **Given** a logged-in user accesses the custom workout section
- **When** exercises are selected
- **And** repetitions and sets are configured
- **And** the workout is saved
- **Then** the custom workout shall be stored in the database
- **And** be available for future use

AC-5: Completing a Workout Session

- **Given** a logged-in user starts a workout
- **When** the workout is completed
- **And** the session is marked as finished
- **Then** the workout session shall be recorded
- **And** the completion date and duration shall be saved
- **And** the session shall appear in workout history

AC-6: Viewing Profile and Workout History

- **Given** a logged-in user accesses the profile page
- **When** the page is loaded

- ➢ **Then** the system shall display basic account information
- ➢ **And** show total completed workouts
- ➢ **And** list past workout sessions

AC-7: Admin Managing Exercises

- ➢ **Given** an administrator is logged into the system
- ➢ **When** a new exercise is added with valid details
- ➢ **Then** the exercise shall be stored in the database
- ➢ **And** become visible in the exercise library

- ➢ **Given** an administrator edits or deletes an exercise
- ➢ **Then** the changes shall be reflected immediately in the system

3.0 NON-FUNCTIONAL REQUIREMENTS

3.1 Performance Requirements

- API response time shall not exceed **2 seconds** for 95% of requests under normal load.
- Web pages shall load within **3 seconds** on a stable broadband connection.
- Mobile application cold start time shall not exceed **3 seconds**.
- The system shall support at least **50–100 concurrent users**.
- Database queries shall complete within **500 milliseconds** under normal conditions.
- The system shall support efficient storage and retrieval of exercise video data stored as binary large objects (BLOB) in the database.

3.2 Security Requirements

- All communications between client and server shall use **HTTPS**.
- User authentication shall be implemented using **JWT tokens**.
- Passwords shall be hashed using **bcrypt** before storage in the database.
- Role-based access control shall restrict admin-only operations.
- The system shall validate user inputs to prevent unauthorized data manipulation and security vulnerabilities.
- The system shall validate all user inputs before processing.

3.3 Compatibility Requirements

- **Web Browsers:** Chrome, Firefox, Safari, Edge (latest 2 versions)
- **Android:** API Level 24+ (Android 7.0+)

- **Screen Sizes:** Mobile (360px+), Tablet (768px+), Desktop (1024px+)
- **Operating Systems:** Windows 10+, macOS 10.15+, Linux Ubuntu 20.04+

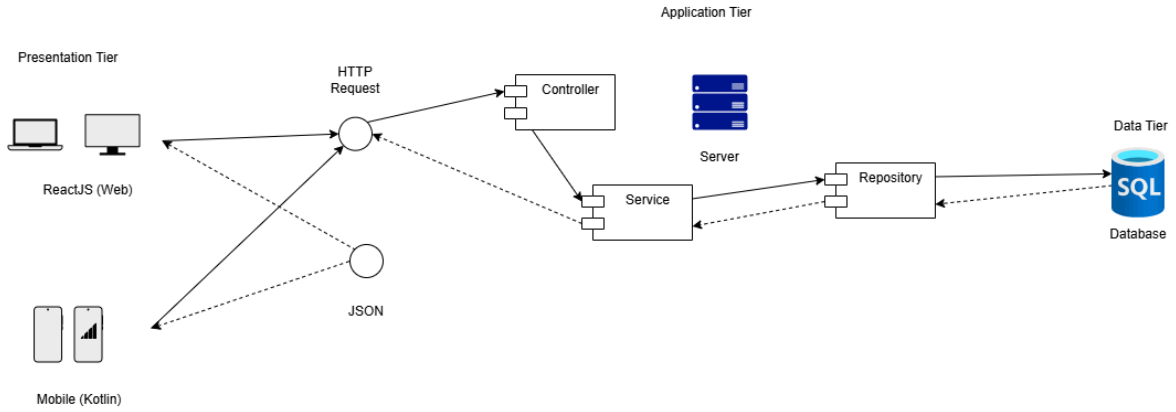
3.4 Usability Requirements

- A new user shall be able to complete their first workout within **5 minutes** of registration.
- Navigation shall remain consistent across all screens.
- Error messages shall clearly describe the issue and provide corrective guidance.

- Buttons and interactive elements on mobile shall meet minimum touch size standards (44x44px).
- Forms shall provide immediate validation feedback.

4.0 SYSTEM ARCHITECTURE

4.1 Component Diagram



Technology Stack:

- **Backend:** Java 17, Spring Boot 3.x, Spring Security, Spring Data JPA
- **Database:** MySQL
- **Web Frontend:** ReactJS, CSS
- **Mobile:** Kotlin
- **Build Tools:** Maven (Backend), npm (Web), Gradle (Android)
- **Deployment:** Railway/Heroku (Backend), Vercel/Netlify (Web), APK (Mobile)

5.0 API CONTRACT & COMMUNICATION

5.1 API Standards

Base URL	https://localhost:8080/api/v1
Format	JSON for all requests/responses
Authentication	Bearer token (JWT) in Authorization header

Response Structure	{
	"success": boolean,
	"data": object|null,
	"error": {
	"code": string,
	"message": string,
	"details": object|null
	},
	"timestamp": string
	}

5.2 Endpoint Specifications

Authentication Endpoints

User Registration

Description	Creates a new user account in the system
API URL	/api/auth/register
HTTP Request Method	POST
Format	JSON for all requests/responses
Authentication	None

Request Payload	{
	"firstName": "<firstName>",
	"lastName": "<lastName>",
	"email": "<email>",
	"password": "<password>"
	}

Response Structure	{
	"success": true,
	"data": {
	"user": {
	"id": "<id>",
	"firstName": "<firstName>",
	"lastName": "<lastName>",
	"email": "<email>",

	"role": "<role>"
	},
	"accessToken": "<jwtAccessToken>"
	},
	"error": null,
	"timestamp": "<timestamp>"
	}
	
	{
	"success": false,
	"data": null,
	"error": {
	"code": "VALID-001",
	"message": "Validation failed",
	"details": {
	"email": "Email already exists"
	}
	},
	"timestamp": "<timestamp>"
	}

User Login

Description	User login in the system
API URL	/api/auth/login
HTTP Method	POST
Format	JSON for all requests/responses
Authentication	None

Request Payload	{
	"email": "<email>",
	"password": "<password>"
	}

Response Structure	{
	"success": true,
	"data": {
	"user": {
	"id": "<id>",
	"firstName": "<firstName>",
	"lastName": "<lastName>",
	"email": "<email>",
	"role": "<role>"
	},
	"accessToken": "<jwtAccessToken>"

	<pre>},
	"error": null,
	"timestamp": "<timestamp>"
	}
	
	{
	"success": false,
	"data": null,
	"error": {
	"code": "AUTH-001",
	"message": "Invalid credentials",
	"details": "Email or password is incorrect"
	},
	"timestamp": "<timestamp>"
	}</pre>

5.3 Error Handling

HTTP Status Codes

- 200 OK - Successful request
- 201 Created - Resource created
- 400 Bad Request - Invalid input
- 401 Unauthorized - Authentication required/failed
- 403 Forbidden - Insufficient permissions
- 404 Not Found - Resource doesn't exist
- 409 Conflict - Duplicate resource
- 500 Internal Server Error - Server error

Error Code Examples

Example 1	<pre>json
	{
	"success": false,
	"data": null,
	"error": {
	"code": "AUTH-001",
	"message": "Invalid credentials",
	"details": "Email or password is incorrect"
	},
	"timestamp": "2024-01-28T10:30:00Z"</pre>

	}

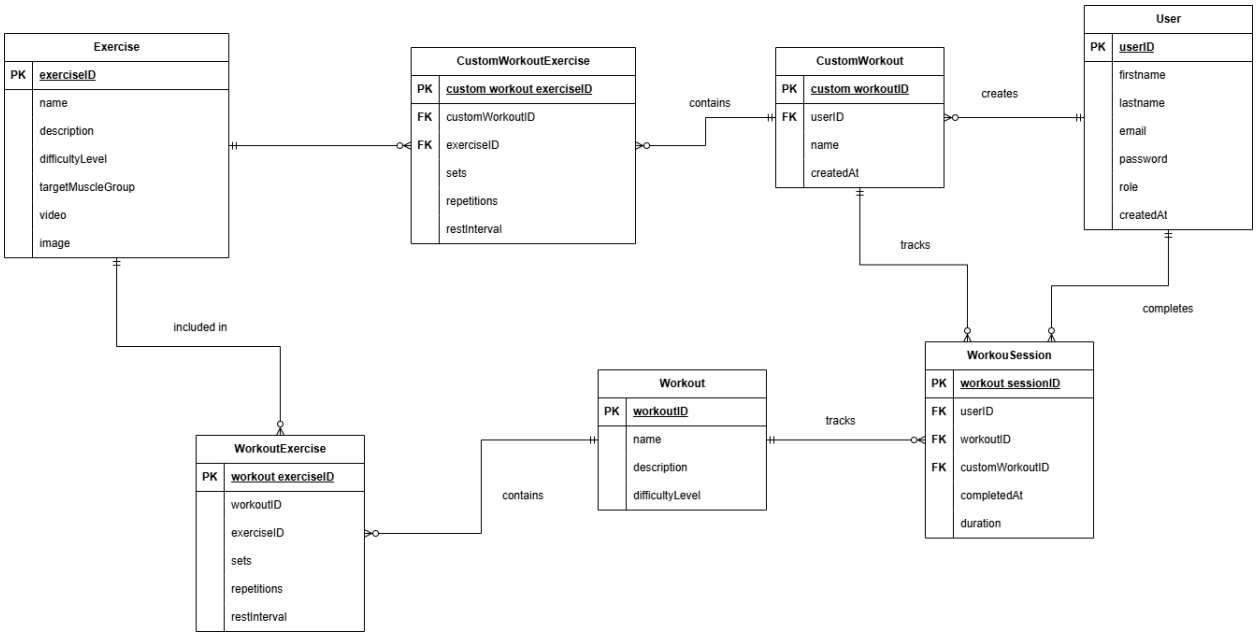
Example 2	{
	"success": false,
	"data": null,
	"error": {
	"code": "VALID-001",
	"message": "Validation failed",
	"details": {
	"email": "Email is required",
	"password": "Must be at least 8 characters"
	}
	},
	"timestamp": "2024-01-28T10:30:00Z"
	}

Common Error Codes

- AUTH-001: Invalid credentials
- AUTH-002: Token expired
- AUTH-003: Insufficient permissions
- VALID-001: Validation failed
- VALID-002: Missing required field
- VALID-003: Invalid input format
- DB-001: Resource not found
- DB-002: Duplicate entry
- DB-003: Database operation failed
- SYSTEM-001: Internal server error
- SYSTEM-002: Unexpected system failure
- WORKOUT-001: Workout not found
- WORKOUT-002: Workout already completed
- SESSION-001: Invalid workout session state
- SESSION-002: Session not found

6.0 DATABASE DESIGN

6.1 Entity Relationship Diagram



Detailed Relationships:

- **One-to-Many:** User → CustomWorkouts
  (Each user can create multiple custom workouts)

- **One-to-Many:** User → WorkoutSessions
  (Each user can have multiple completed workout sessions)

- **One-to-Many:** Workout → WorkoutExercises
  (Each predefined workout contains multiple exercises)

- **One-to-Many:** CustomWorkout → CustomWorkoutExercises
  (Each custom workout contains multiple exercises)

- **Many-to-Many:** Exercise ↔ MuscleGroup
  (Resolved through the exercise_muscle_groups junction table)

- **Many-to-One:** WorkoutExercises → Exercise
  (Workout exercises reference exercises)

- **Many-to-One:** CustomWorkoutExercises → Exercise
  (Custom workout exercises reference exercises)

- **Many-to-One:** WorkoutSessions → User
  (Each workout session belongs to one user)

- **Many-to-One:** WorkoutSessions → Workout
  (A session may reference a predefined workout)

- **Many-to-One:** WorkoutSessions → CustomWorkout
  (A session may reference a custom workout)

Key Tables:

1. **User** - User accounts and authentication
2. **Exercise** - Bodyweight exercise information
3. **MuscleGroup -** List of available muscle groups
4. **ExerciseMuscleGroup** - Mapping table between exercises and muscle groups
5. **Workout** - Predefined workout programs
6. **WorkoutExercises** - Exercises included in predefined workouts
7. **CustomWorkouts** - User-created workout routines
8. **CustomWorkoutExercises** - Exercises included in custom workouts
9. **WorkoutSessions** - Completed workout records

Table Structure Summary:

- **User** : user_id, firstname, lastname, email, password, role, created_at

- **Exercise** : exercise_id, name, description, difficulty_level, target_muscle_group, video, image

- **MuscleGroup:** muscle_group_id, name

- **ExerciseMuscleGroup** : exercise_muscle_group_id,muscle_group_id, exercise_id

- **Workout** : workout_id, name, description, difficulty_level

- **WorkoutExercises** : workout_exercise_id, workout_id, exercise_id, sets, repetitions, rest_interval

- **CustomWorkouts** : custom_workout_id, user_id, name, created_at

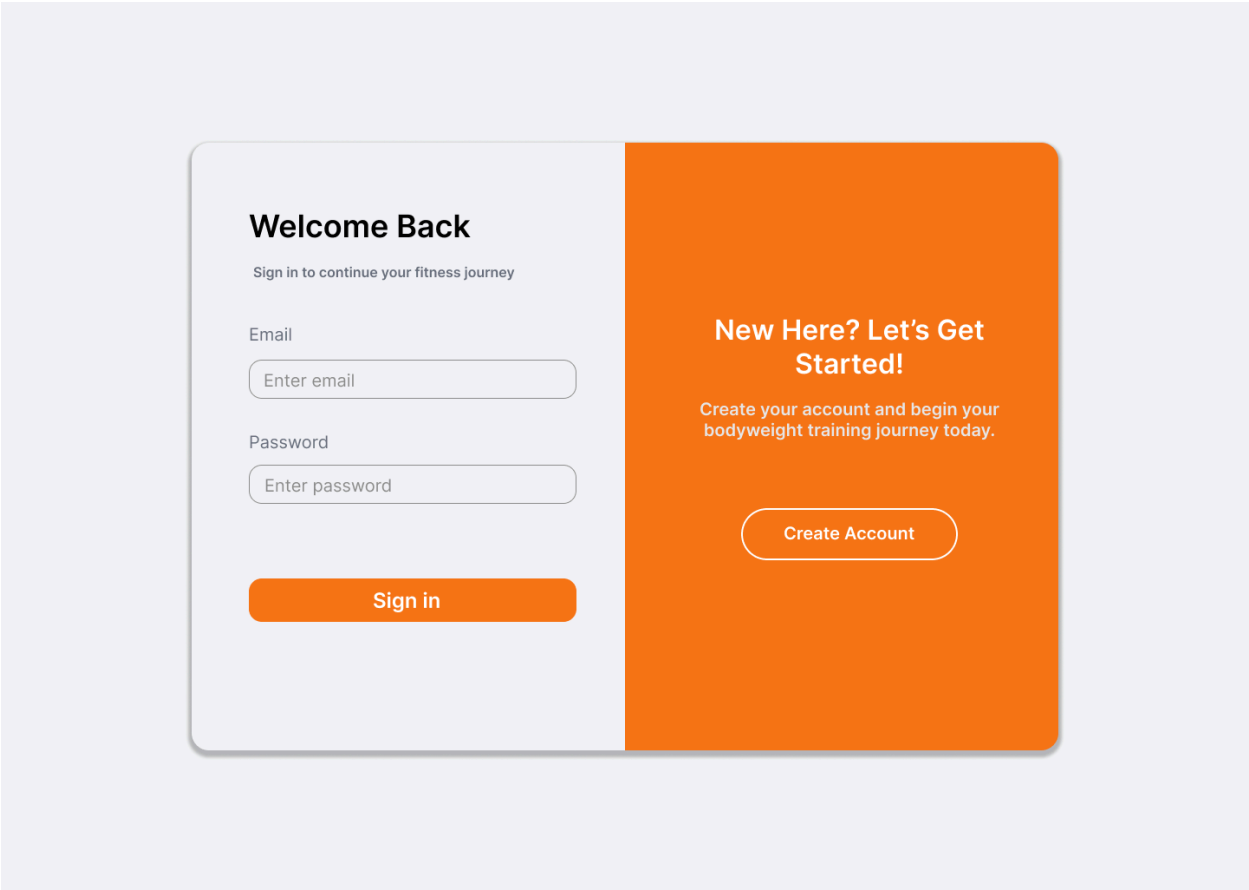
- **CustomWorkoutExercises** : custom_workout_exercises_id, custom_workout_id, exercise_id, sets, repetitions, rest_interval

- **WorkoutSessions** : workout_session_id, user_id, workout_id (nullable), custom_workout_id (nullable), completed_at, duration

7.0 UI/UX DESIGN

7.1 Web Application Wireframes

Login Page



Header:
None (Authentication-focused layout)

Content (Two-Column Card Layout):

Left Section:

- Title: "Welcome Back"
- Subtitle: "Sign in to continue your fitness journey"

- Email Input Field
- Password Input Field
- “Sign in” Button

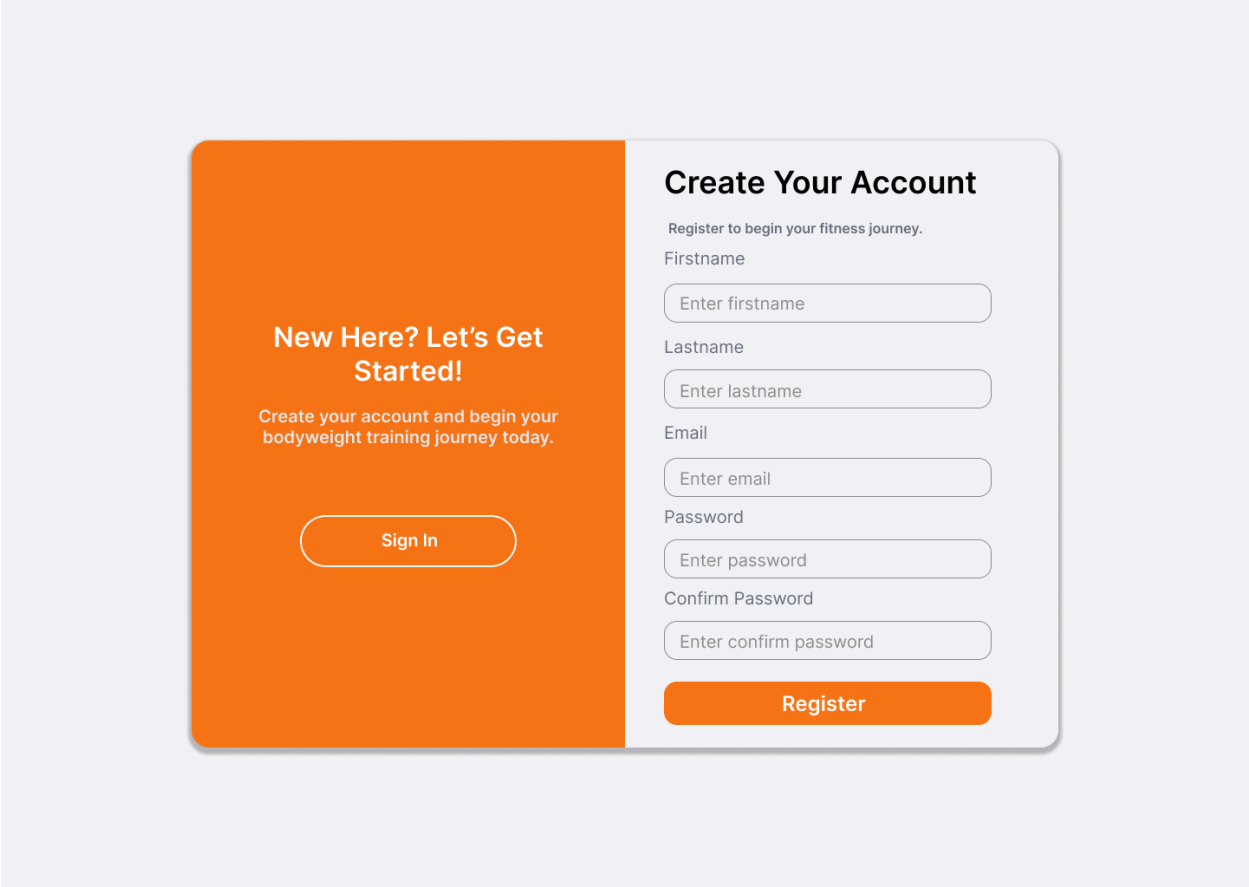
Right Section (Orange Panel):

- Title: “New Here? Let’s Get Started!”
- Subtitle: “Create your account and begin your bodyweight training journey today.”
- “Create Account” Button

Background:
Light gray page background with centered card container.

Footer:
None

Register Page



Header:
None (Authentication-focused layout)

Content (Two-Column Card Layout):

Left Section (Orange Panel):

- Title: "New Here? Let's Get Started!"
- Subtitle encouraging account creation
- "Sign In" Button

Right Section (Form Panel):

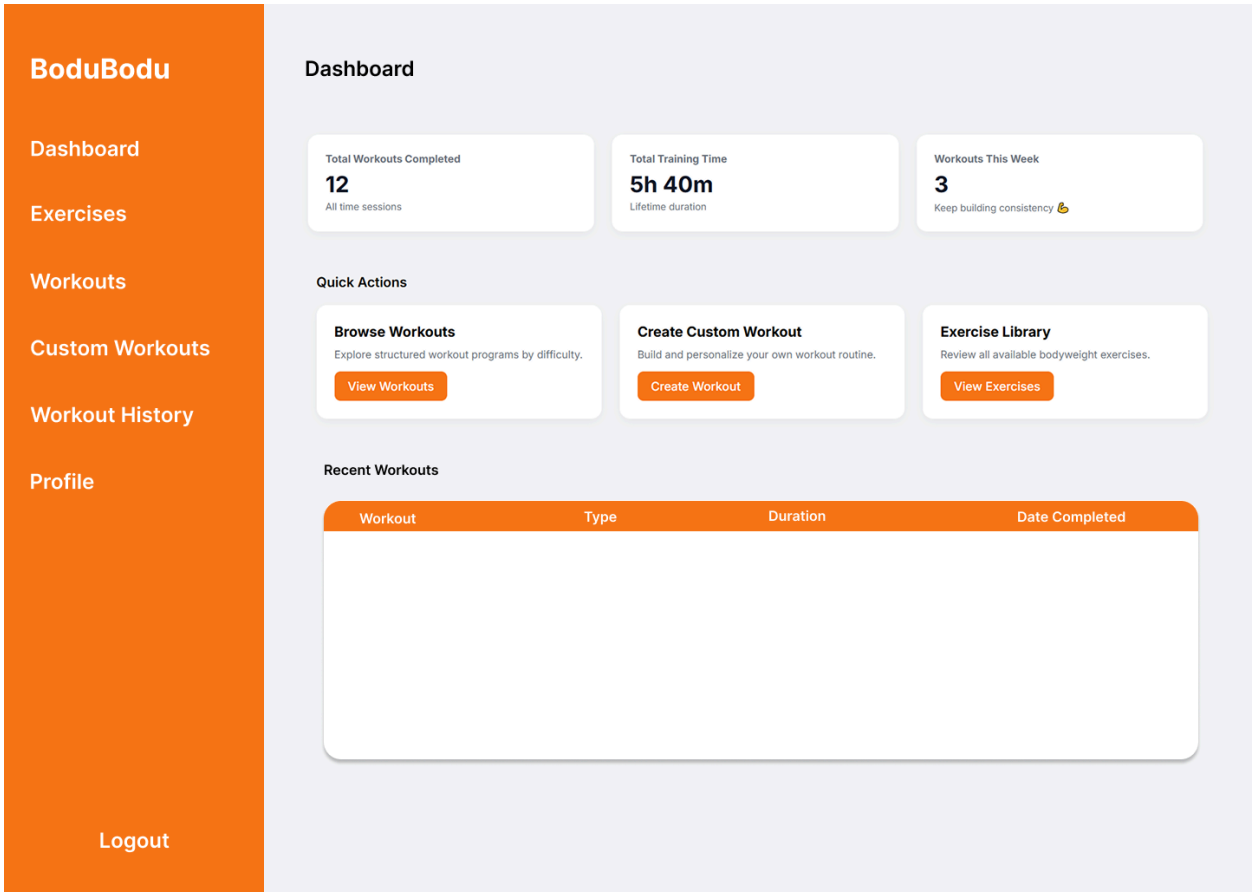
- Title: "Create Your Account"

- Subtitle: “Register to begin your fitness journey.”
- Firstname Input Field
- Lastname Input Field
- Email Input Field
- Password Input Field
- Confirm Password Input Field
- “Register” Button

Background:
Light gray page background with centered card container.

Footer:
None

Dashboard Page



Header:
Top-left: “BoduBodu”

Sidebar Navigation (Left Panel):

- Dashboard
- Exercises
- Workouts
- Custom Workouts
- Workout History
- Profile

- Logout (Bottom)

Content (Main Panel):

Statistics Cards:

- Total Workouts Completed
- Total Training Time
- Workouts This Week

Quick Actions Section:

- Browse Workouts (View Workouts button)
- Create Custom Workout (Create Workout button)
- Exercise Library (View Exercises button)

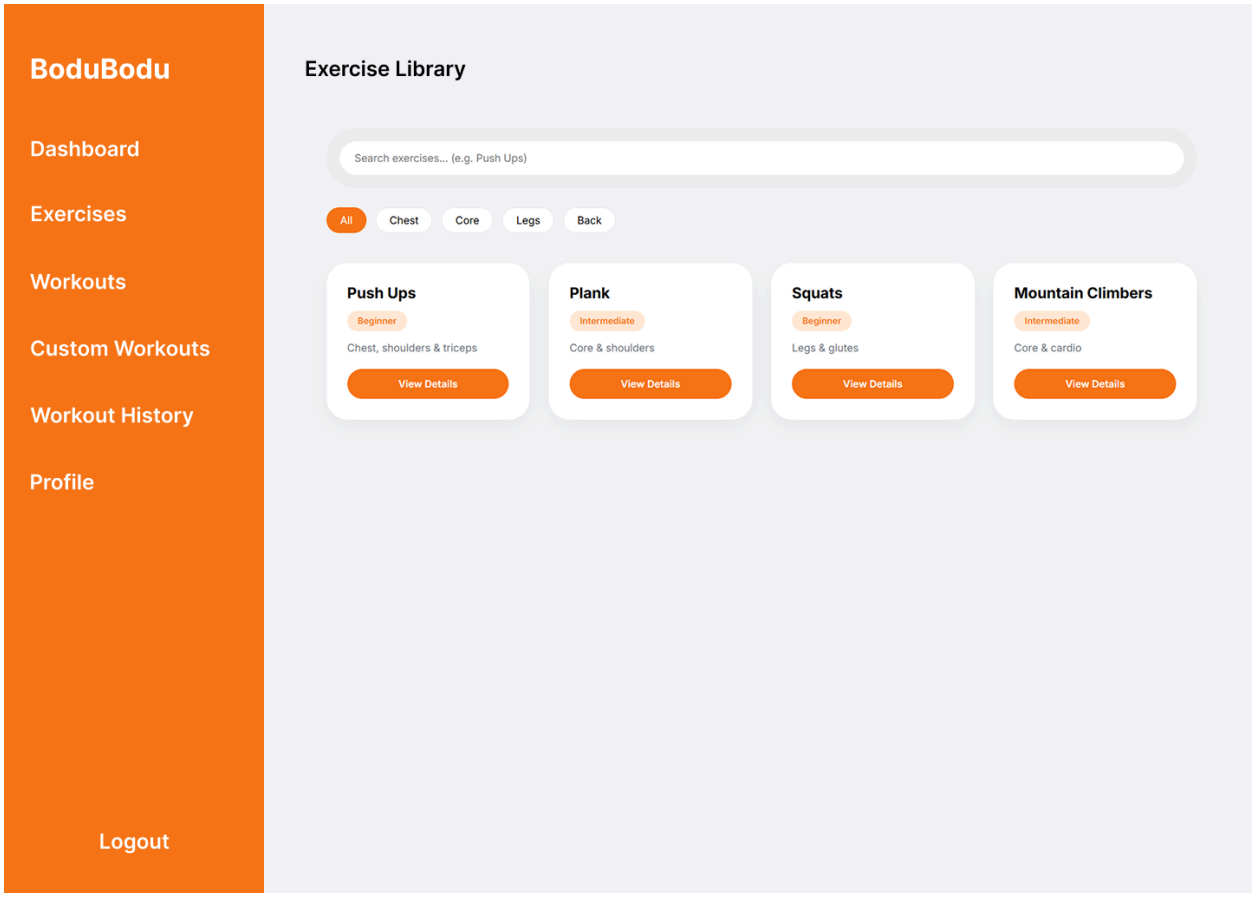
Recent Workouts Table:
Columns:

- Workout
- Type
- Duration
- Date Completed

Background:
White/Light gray main content area with orange sidebar.

Footer:
None

Exercise Library (User view)



Header:
Page Title: “Exercise Library”

Sidebar Navigation:
Same left navigation menu.

Content:

- Search Bar (Search exercises...)
- Category Filters:
    - All
    - Chest
    - Core

    - Legs
    - Back

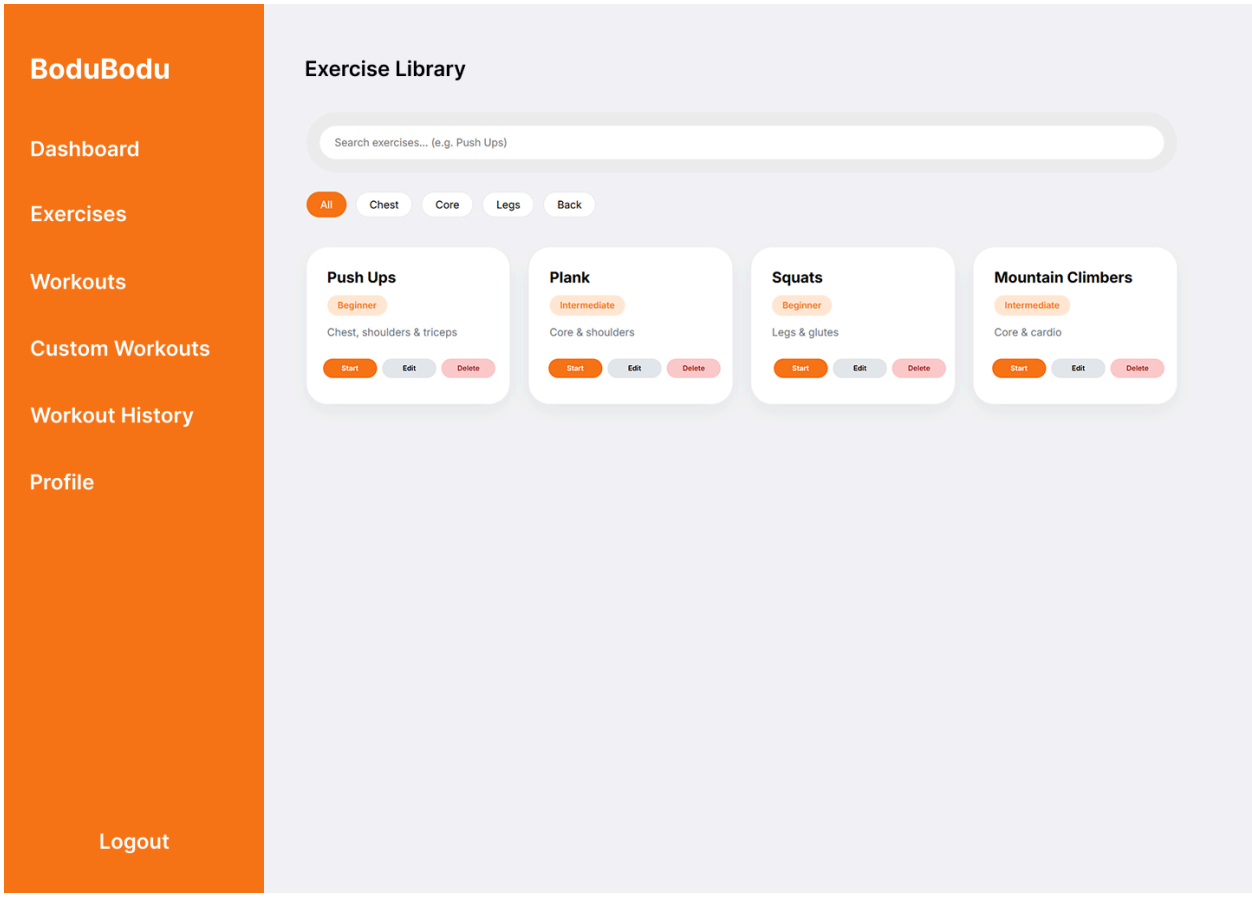
Exercise Cards:
Each card includes:

- Exercise Name (e.g., Push Ups)
- Difficulty Badge (Beginner / Intermediate)
- Target Muscles
- “View Details” Button

Background:
Light gray content background.

Footer:
None

Exercise Library (Admin view)



Header:
Page Title: “Exercise Library”

Sidebar Navigation:
Same left menu.

Content:

- Search Bar
- Category Filters

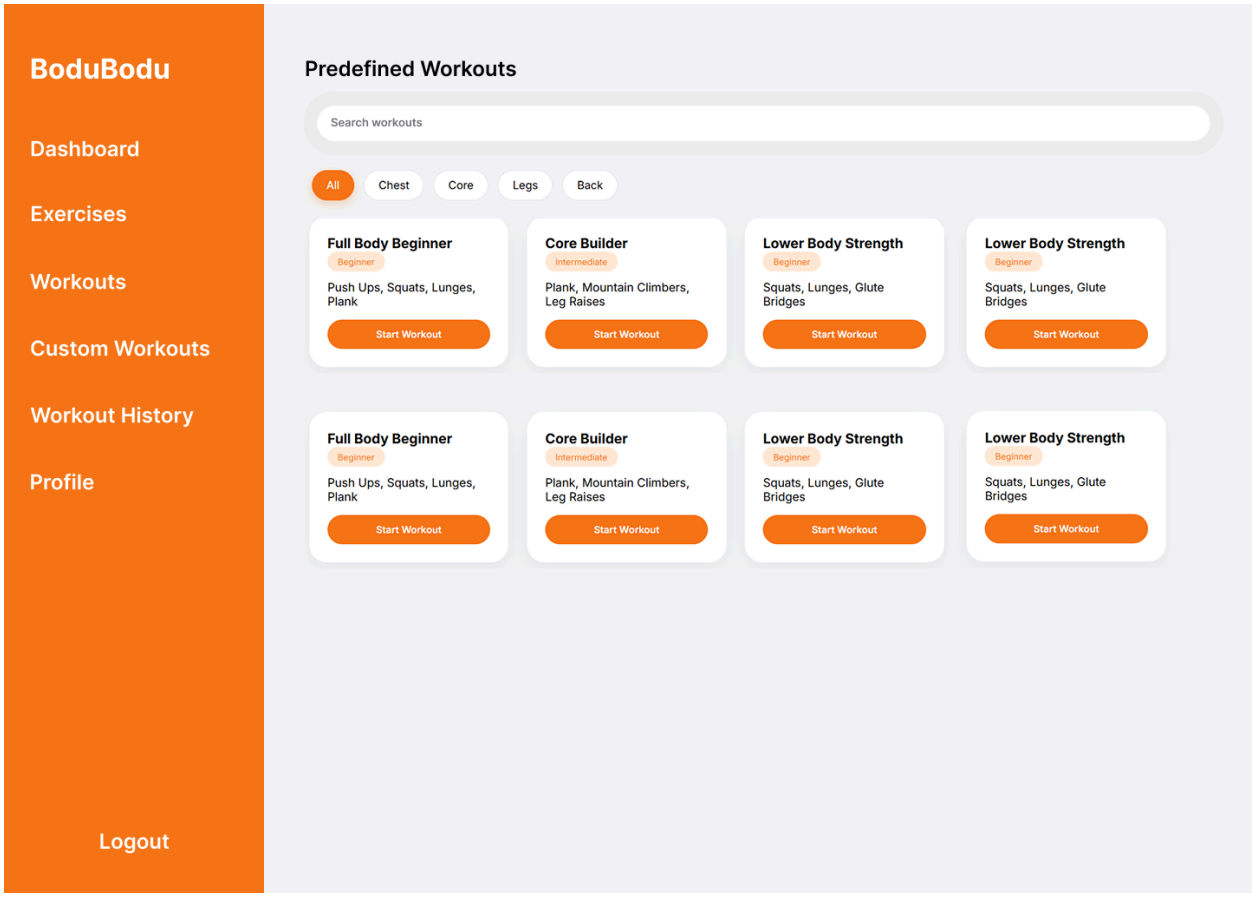
Exercise Cards:
Each card includes:

- Exercise Name
- Difficulty
- Target Muscles
- Action Buttons:
    - Start
    - Edit
    - Delete

Background:
Light gray content background.

Footer:
None

Predefined Workouts (User View)



Header:
Page Title: “Exercise Library”

Sidebar Navigation:
Same left menu.

Content:

- Search Bar
- Category Filters

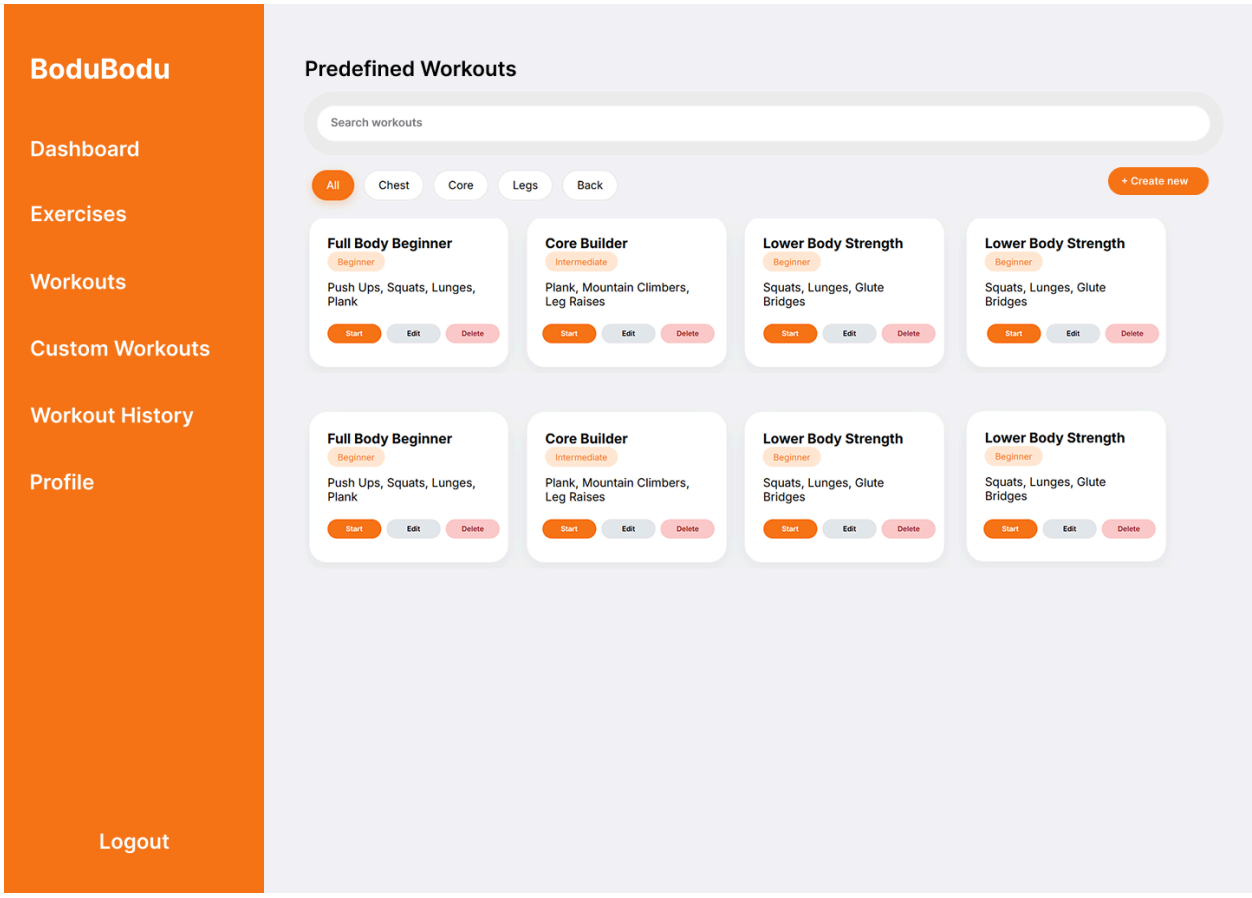
Exercise Cards:
Each card includes:

- Exercise Name
- Difficulty
- Target Muscles
- Action Buttons:
    - Start
    - Edit
    - Delete

Background:
Light gray content background.

Footer:
None

Predefined Workouts (Admin View)



Header:
Page Title: “Predefined Workouts”

Sidebar Navigation:
Same left menu.

Content:

- Search Bar
- Category Filters
- Workout Cards:

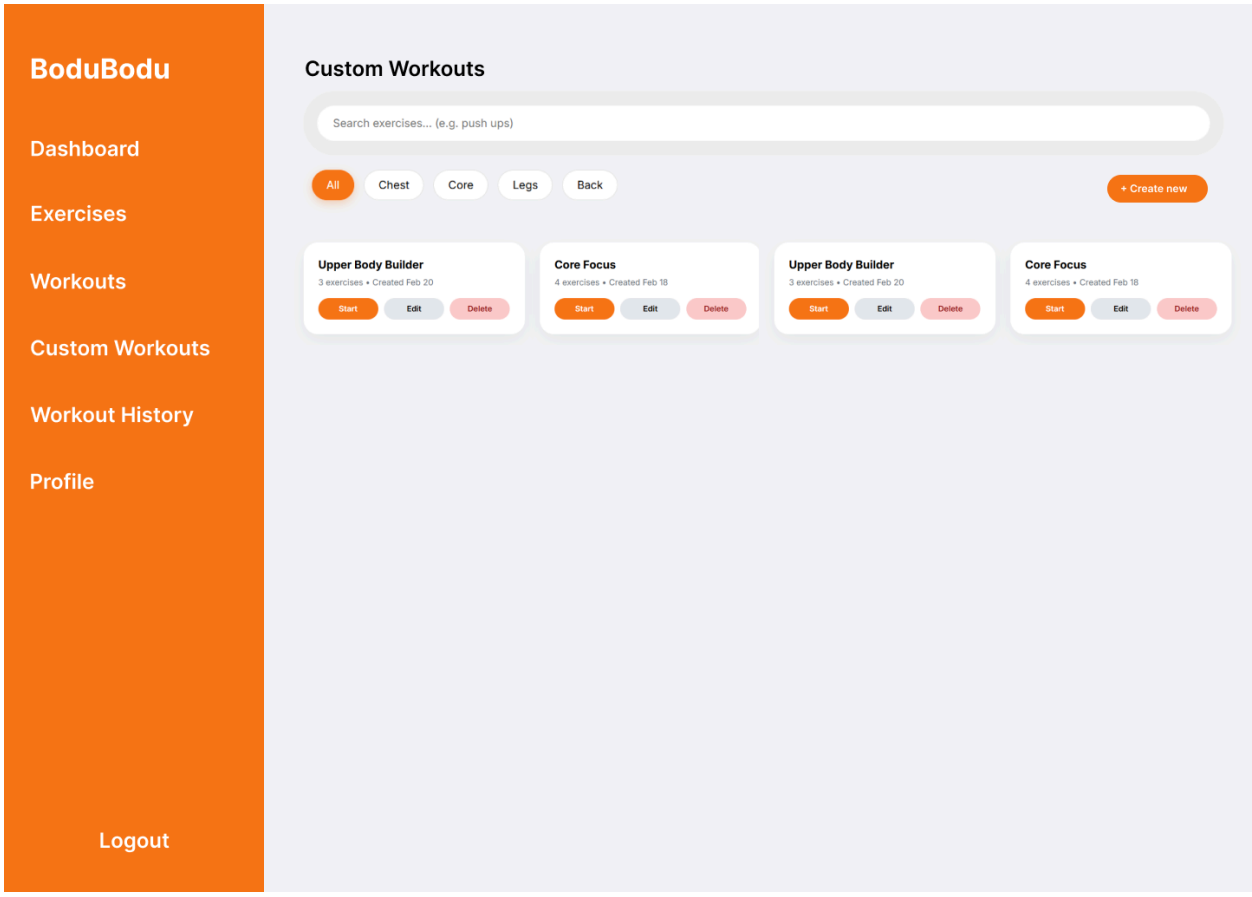
Each card includes:

- Workout Name (e.g., Core Builder)
- Difficulty Badge
- Exercise Preview List
- “Start Workout” Button

Background:
Light gray content background.

Footer:
None

Custom Workout page



Header:
Page Title: “Custom Workouts”

Sidebar Navigation:
Same left menu.

Content:

- Search Bar
- Category Filters
- “+ Create new” Button

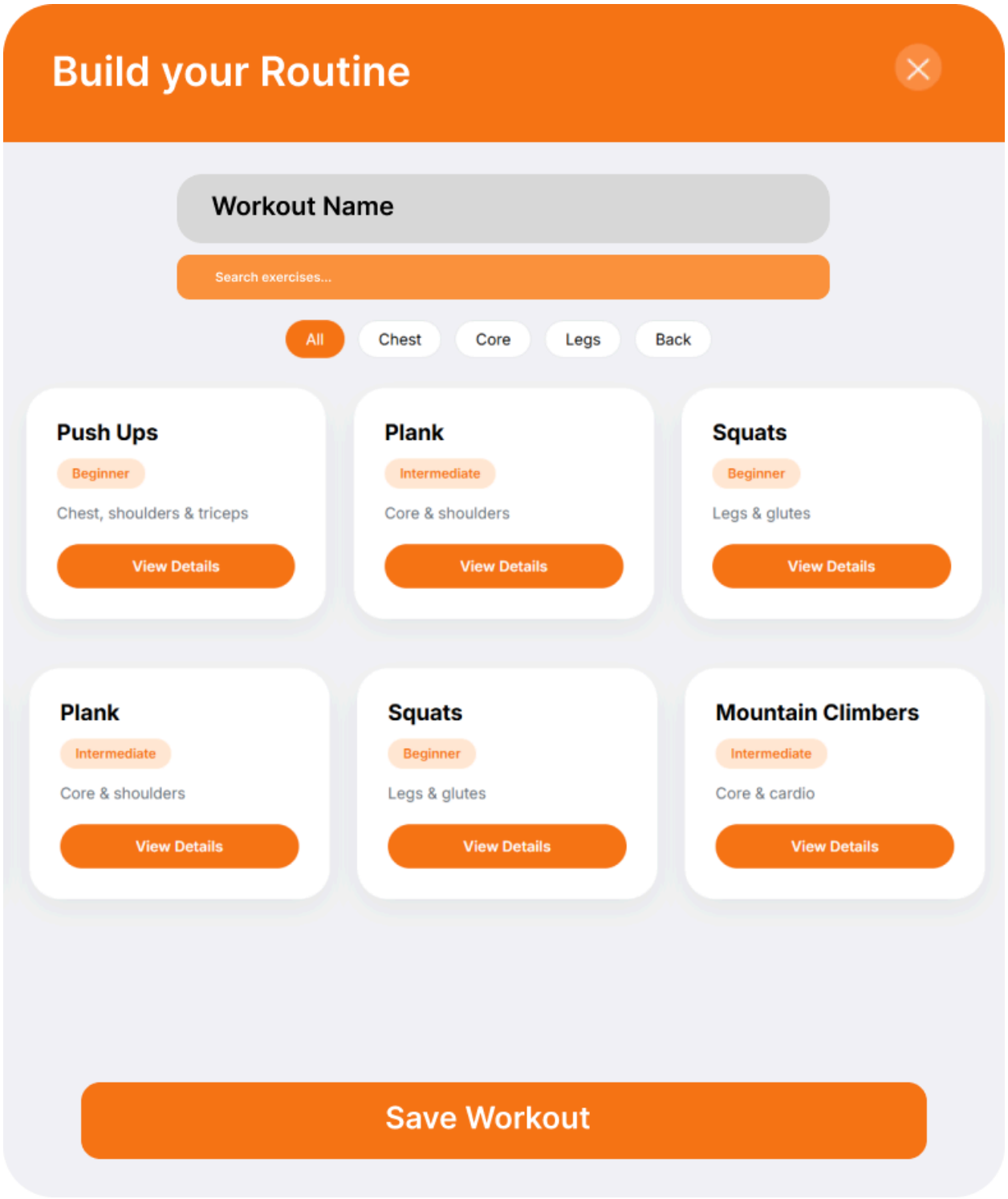
Custom Workout Cards:
Each card includes:

- Workout Name (e.g., Upper Body Builder)
- Number of exercises
- Created Date
- Buttons:
    - Start
    - Edit
    - Delete

Background:
Light gray content background.

Footer:
None

Build your Routine Modal



Header (Modal Top Bar):

- Title: “Build your Routine”
- Close (X) Button

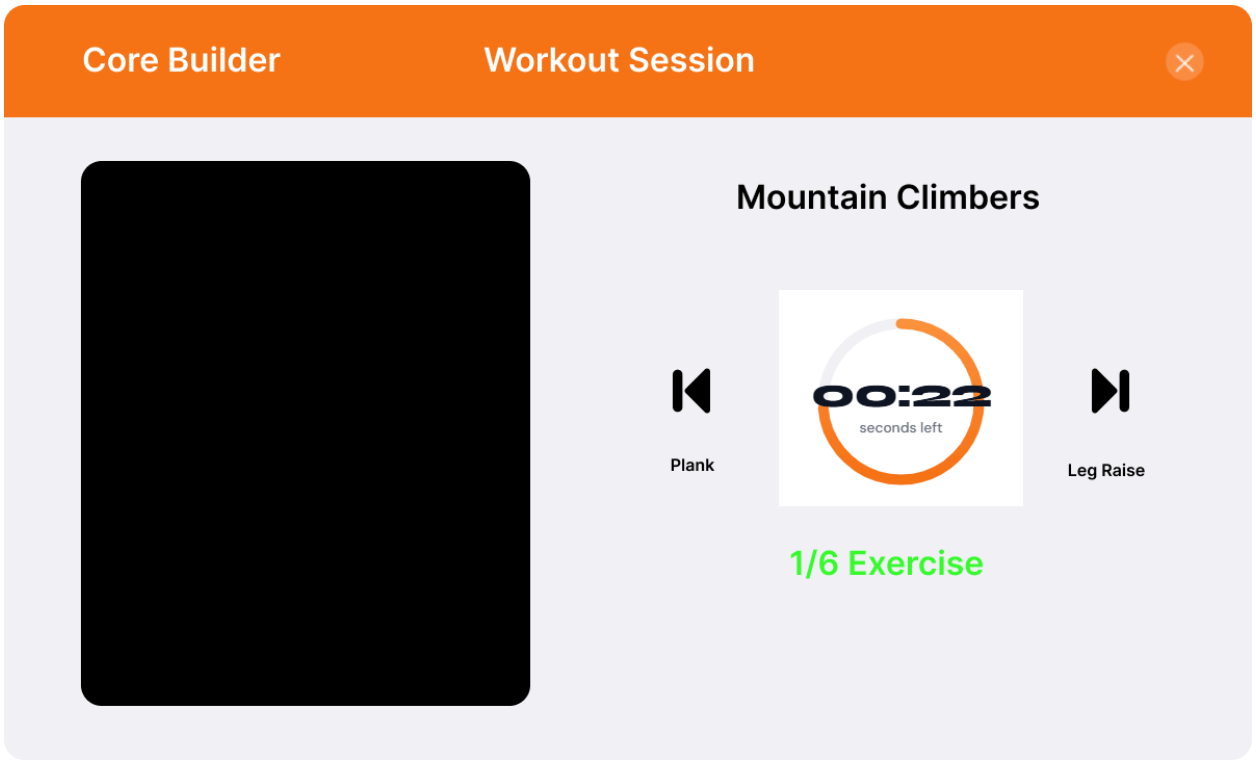
Content (Modal Layout):

- Workout Name Input Field
- Search Exercises Bar
- Category Filters
- Exercise Cards:
    - Exercise Name
    - Difficulty
    - Target Muscles
    - “View Details” Button
- “Save Workout” Button (Bottom)

Background:
Dimmed page background behind modal.

Footer:
None (Modal layout)

Workout Session Modal



Header (Modal Top Bar):

- Workout Name (e.g., Core Builder)
- “Workout Session”
- Close (X) Button

Content:

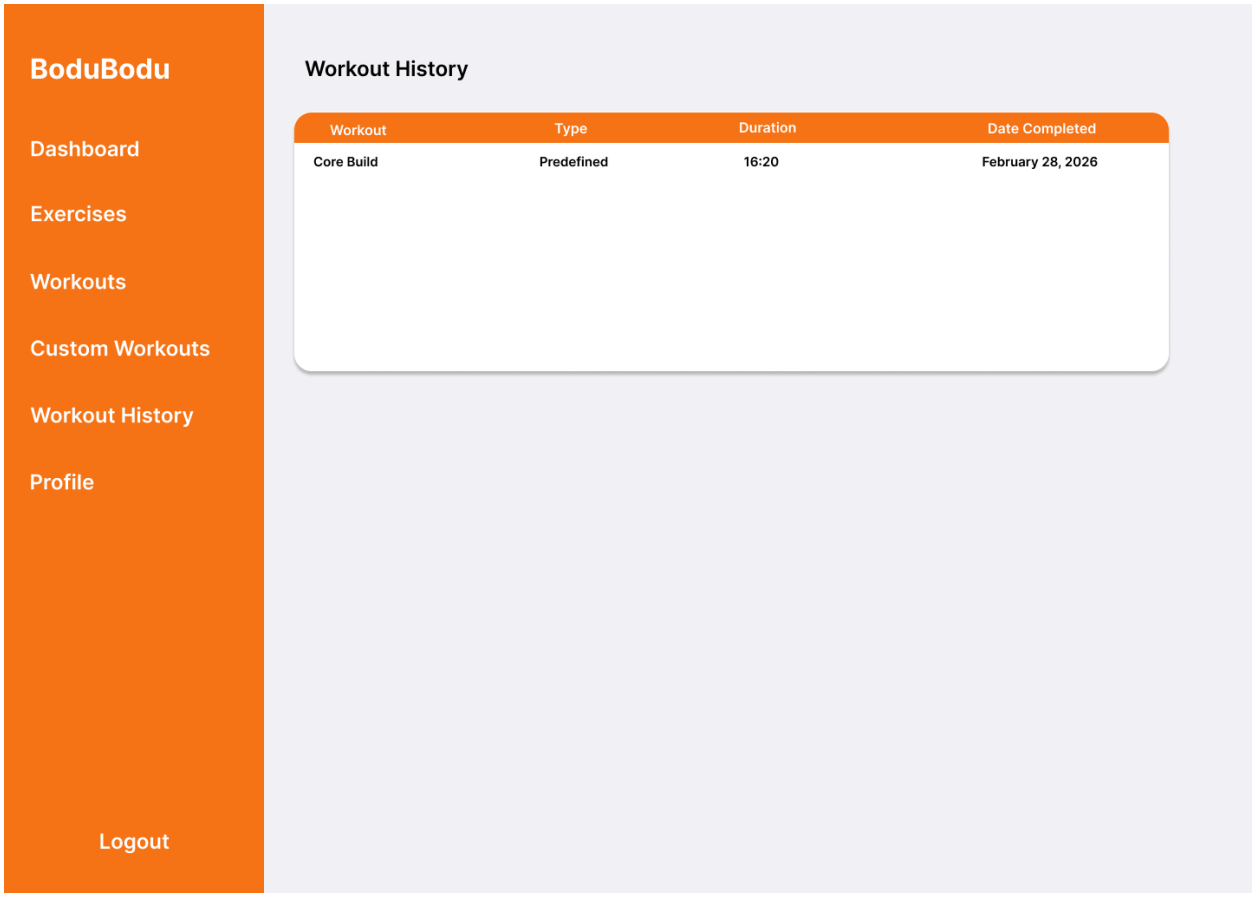
- Exercise Video Section (Left)
- Current Exercise Name (e.g., Mountain Climbers)
- Timer Display (Countdown circle)
- Navigation Controls (Previous / Next)

- Progress Indicator (e.g., 1/6 Exercise)

Background:
Dimmed page background behind modal.

Footer:
None

Workout History Page



Header:
Page Title: “Workout History”

Sidebar Navigation:
Same left menu.

Content:
Table with columns:

- Workout
- Type
- Duration
- Date Completed

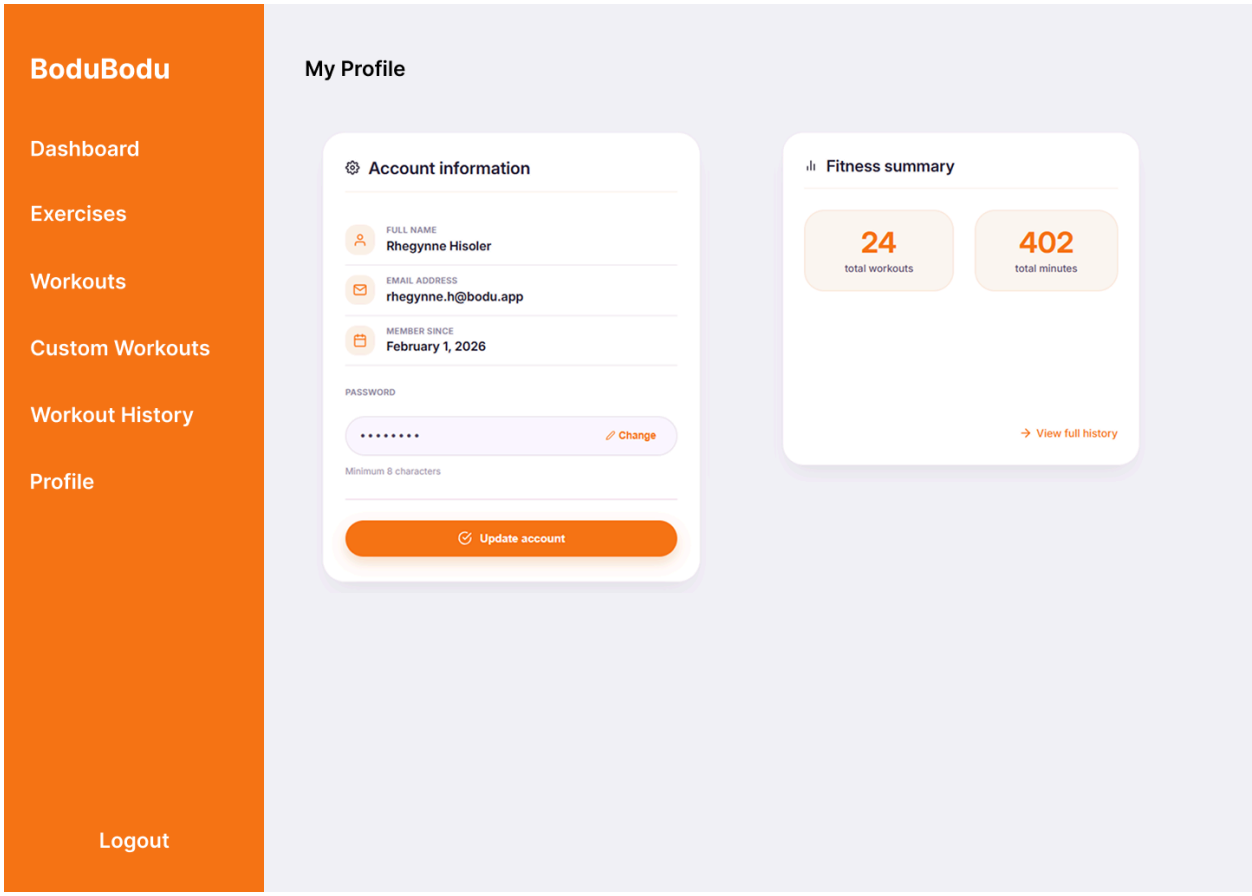
Example Entry:

- Core Build
- Predefined
- 16:20
- February 28, 2026

Background:
Light gray content background.

Footer:
None

Profile Page



Header:
Page Title: “My Profile”

Sidebar Navigation:
Same left menu with Logout at bottom.

Content:

Account Information Card:

- Full Name
- Email Address
- Member Since

- Password (with Change option)
- “Update account” Button

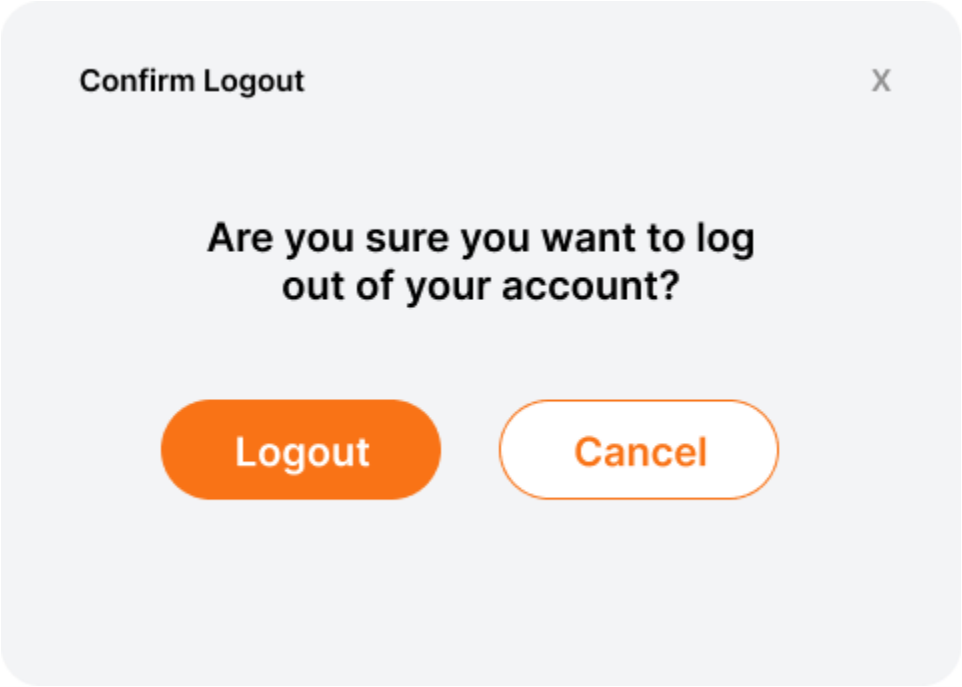
Fitness Summary Card:

- Total Workouts
- Total Minutes
- “View full history” Link

Background:
Light gray content background.

Footer:
None

Logout Modal



Header:
Page Title: “My Profile”

Sidebar Navigation:
Same left menu with Logout at bottom.

Content:

Account Information Card:

- Full Name
- Email Address
- Member Since
- Password (with Change option)
- “Update account” Button

Fitness Summary Card:

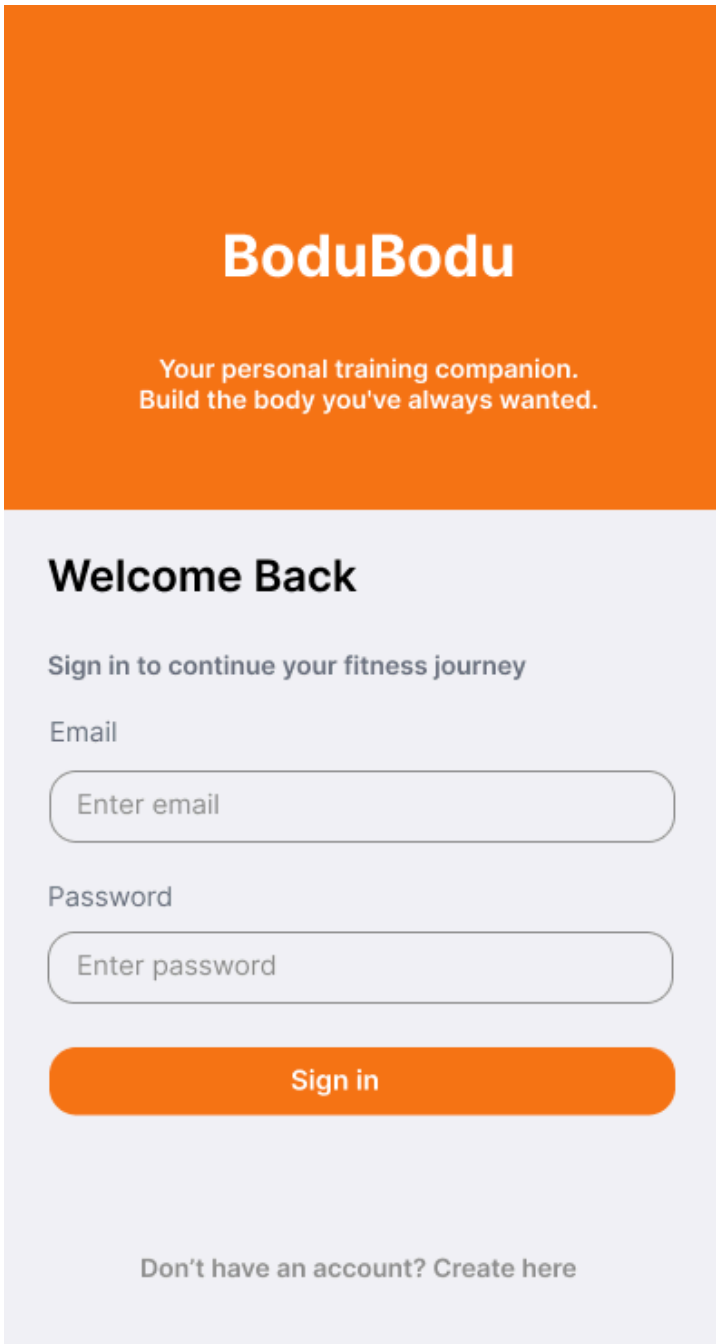
- Total Workouts
- Total Minutes
- “View full history” Link

Background:
Light gray content background.

Footer:
None

7.2 Mobile Application Wireframes

Login Screen



Header:

Orange header section with:

- App Name: **"BoduBodu"**

- Tagline:
  "Your personal training companion.
  Build the body you've always wanted."

Content (Form Layout):

- Title: **"Welcome Back"**
- Subtitle: "Sign in to continue your fitness journey"
- Email Input Field
- Password Input Field
- "Sign in" Button (Primary CTA)
- Text Link: "Don't have an account? Create here"

Background:
Orange top section and light gray lower section.

Footer:
None (Authentication-focused layout)

Register Screen

< Back to login

Create Account

Create your account and begin your bodyweight training journey today.

Firstname

Enter firstname

Lastname

Enter lastname

Email

Enter email

Password

Enter password

Confirm Password

Enter confirm password

Register

Header:

Top-left text link: “< Back to login”

Large Title: **“Create Account”**

Subtitle: “Create your account and begin your bodyweight training journey today.”

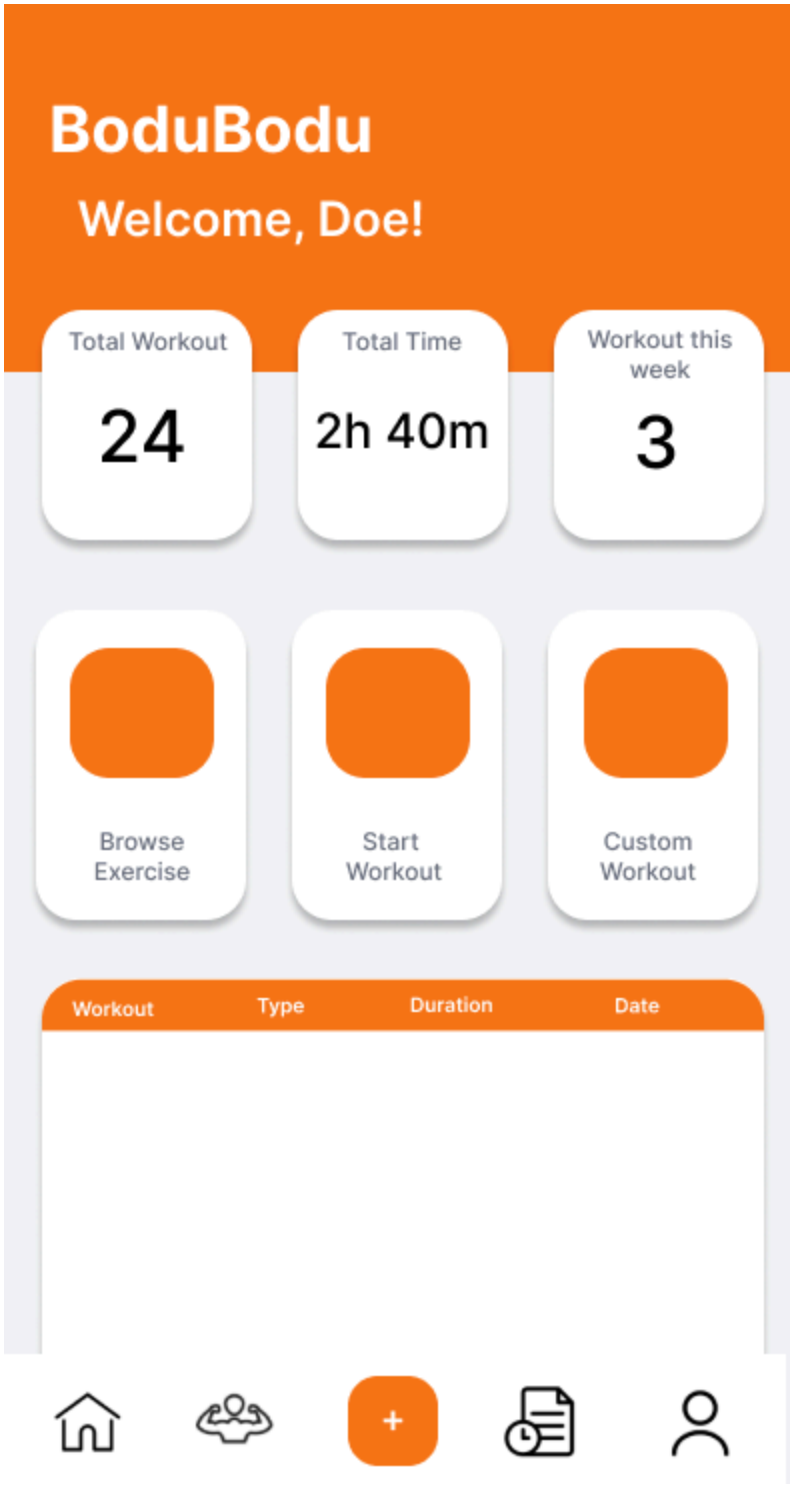
Content (Vertical Form Layout):

- Firstname Input Field
- Lastname Input Field
- Email Input Field
- Password Input Field
- Confirm Password Input Field
- “Register” Button (Primary CTA)

Background:
Orange header section with light gray form section.

Footer:
None

Dashboard/Home Screen



Header:

App Title: **"BoduBodu"**

Greeting: "Welcome, Doe!"

Content (Scrollable Layout):

Statistics Cards:

- Total Workout – 24
- Total Time – 2h 40m
- Workout this week – 3

Quick Action Cards:

- Browse Exercise
- Start Workout
- Custom Workout

Recent Workout Table Section:

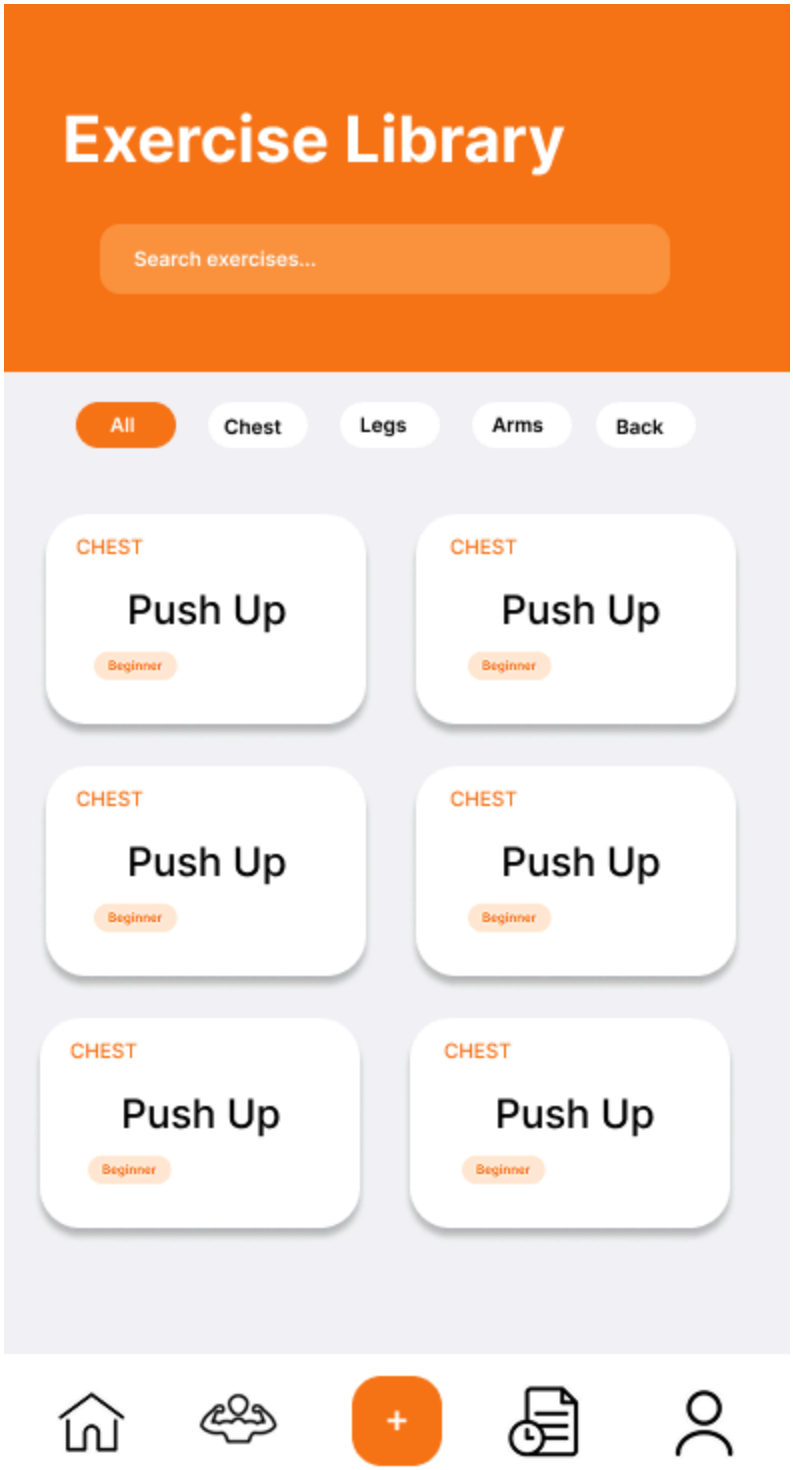
- Columns: Workout | Type | Duration | Date

Background:
Orange header section with light gray content area.

Footer:
Bottom Navigation Bar:

- Home
- Exercise
- Add button (create custom workout)
- History
- Profile

Exercise Library Screen



Header:

Title: **"Exercise Library"**

Content:

- Search Bar (“Search exercises…”)
- Category Filter Chips:
    - All
    - Chest
    - Legs
    - Arms
    - Back

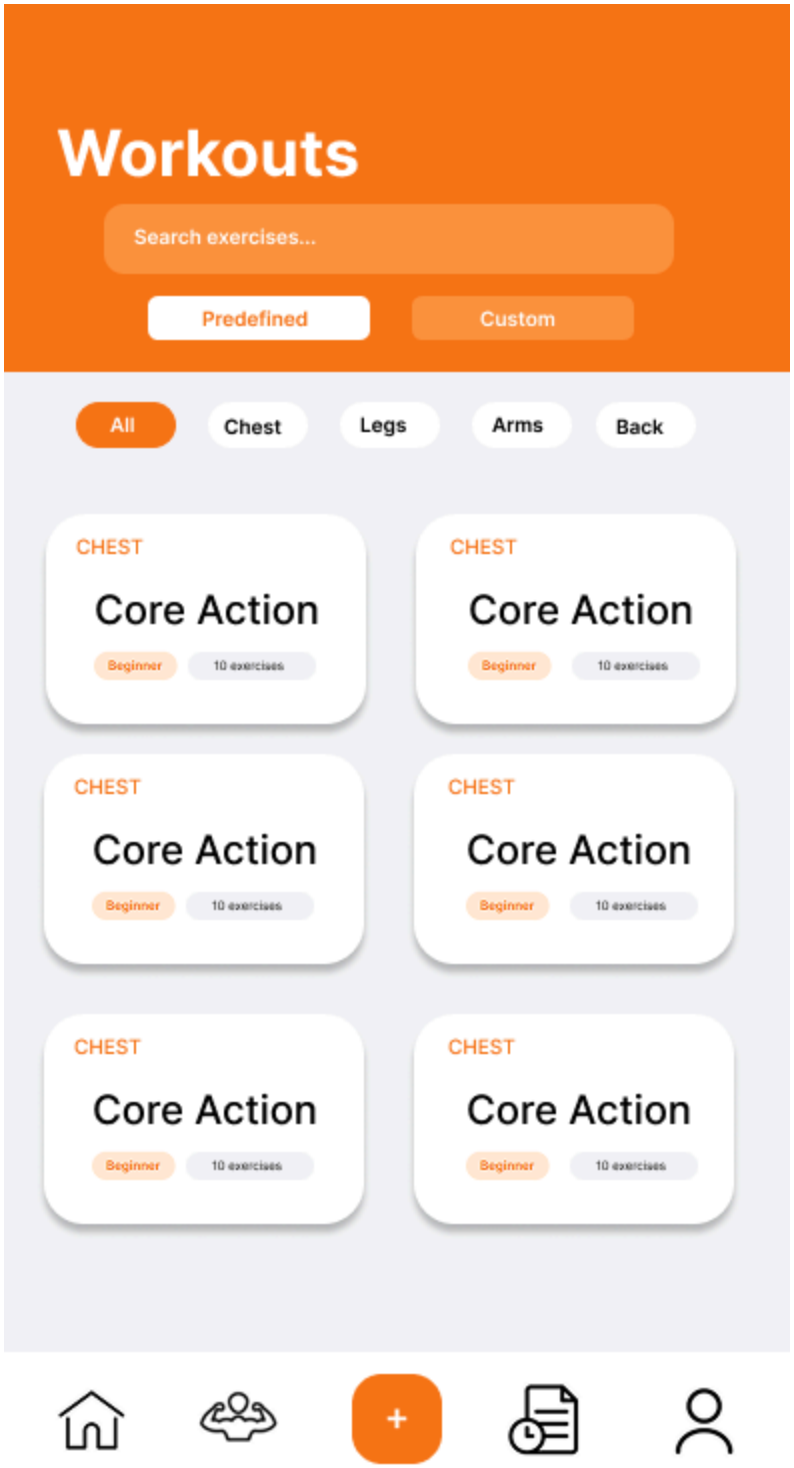
Exercise Cards (Grid Layout):
Each card includes:

- Muscle Tag (e.g., CHEST)
- Exercise Name (e.g., Push Up)
- Difficulty Badge (Beginner)

Background:
Orange header with light gray content.

Footer:
Bottom Navigation Bar

Workout Screen



Header:

Title: **"Workouts"**

Content:

- Search Bar
- Toggle Buttons:
    - Predefined (Active)
    - Custom
- Category Filters (All, Chest, Legs, Arms, Back)

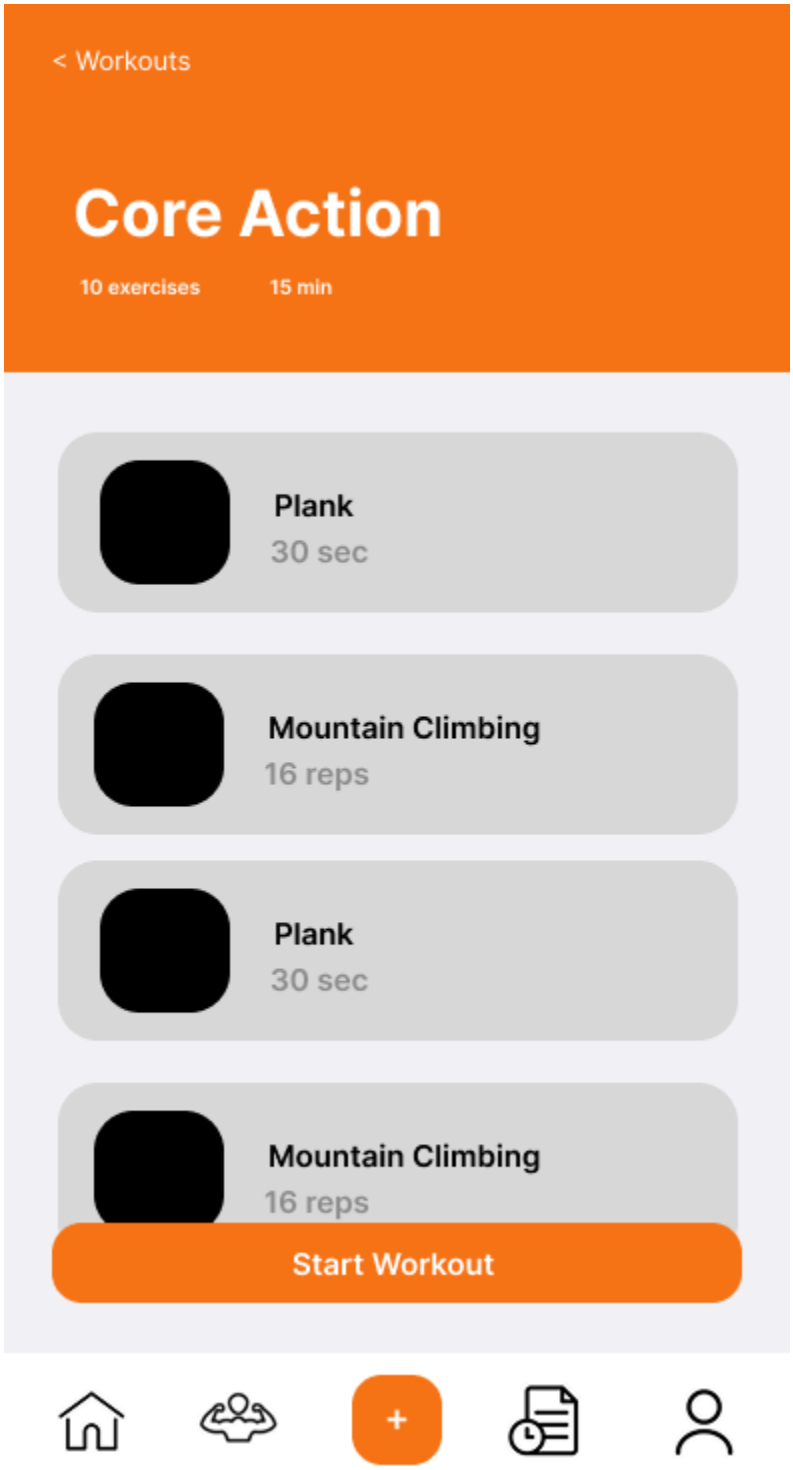
Workout Cards:
Each card shows:

- Muscle Tag (e.g., CHEST)
- Workout Name (e.g., Core Action)
- Difficulty Badge (Beginner)
- Number of exercises (10 exercises)

Background:
Orange header + gray content.

Footer:
Bottom Navigation Bar

Workout Details Screen



Header:
Back Button: “< Workouts”
Workout Title: **“Core Action”**
Details:

- 10 exercises
- 15 min

Content (Exercise List):

- Plank – 30 sec
- Mountain Climbing – 16 reps
- Additional exercises listed vertically

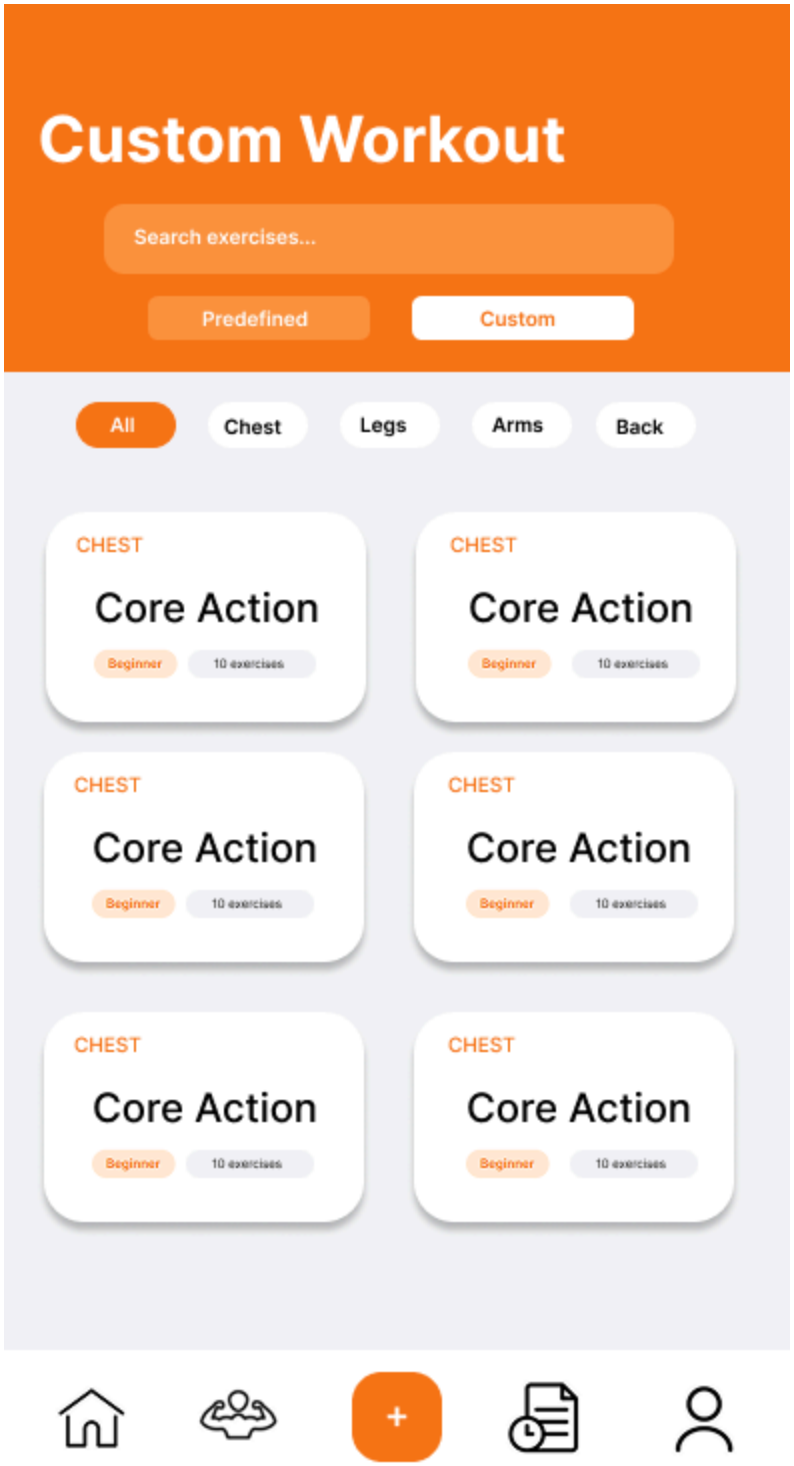
CTA Button:

- “Start Workout” (Bottom Primary Button)

Background:
Orange header with light gray content.

Footer:
Bottom Navigation Bar

Custom Workout Screen



Header:

"< Custom Workouts"

Workout Name (Editable with Pencil Icon)

Workout Summary:

- 02 exercises
- 00 min
- Save Button

Exercise List:
Each item includes:

- Exercise Thumbnail
- Exercise Name (e.g., Plank)
- Duration/Reps
- Edit Icon
- Delete Icon

Floating Action Button:

- "+" (Add Exercise)

Bottom Button:

- "Start Workout"

Background:
Orange header + gray content.

Footer:
Bottom Navigation Bar

Workout History Screen

Header:
Title: **"Workout History"**

Content:
Large card container with table columns:

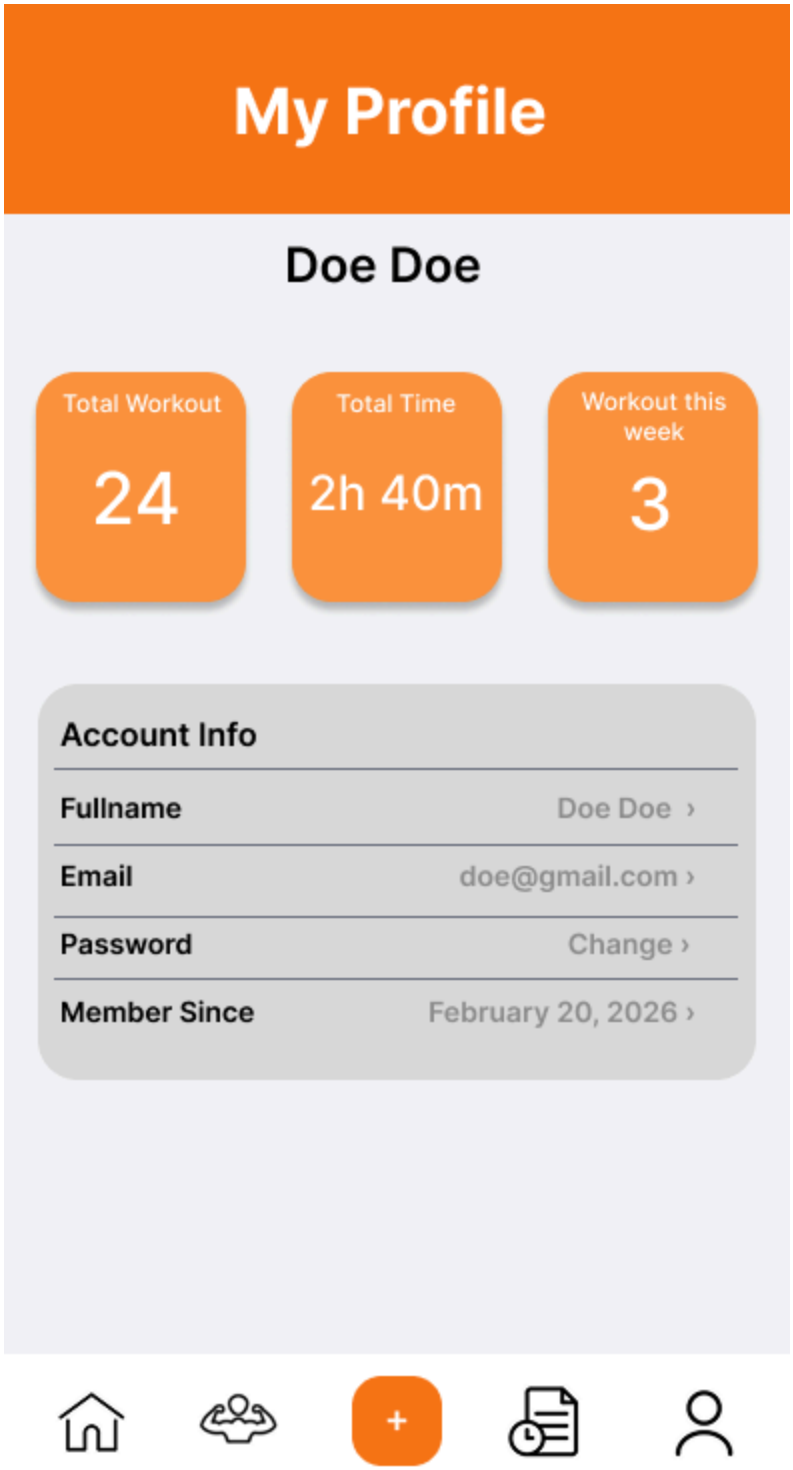
- Workout
- Type
- Duration
- Date

(Currently empty state)

Background:
Orange header + gray content.

Footer:
Bottom Navigation Bar

Profile Screen



Header:

Title: **"My Profile"**

Content (Scrollable Layout):

User Information Section:

- Display Name: Doe Doe

Statistics Cards:

- Total Workout
- Total Time
- Workout this week

Account Info Section:

- Fullname
- Email
- Password (Change option)
- Member Since – February 20, 2026

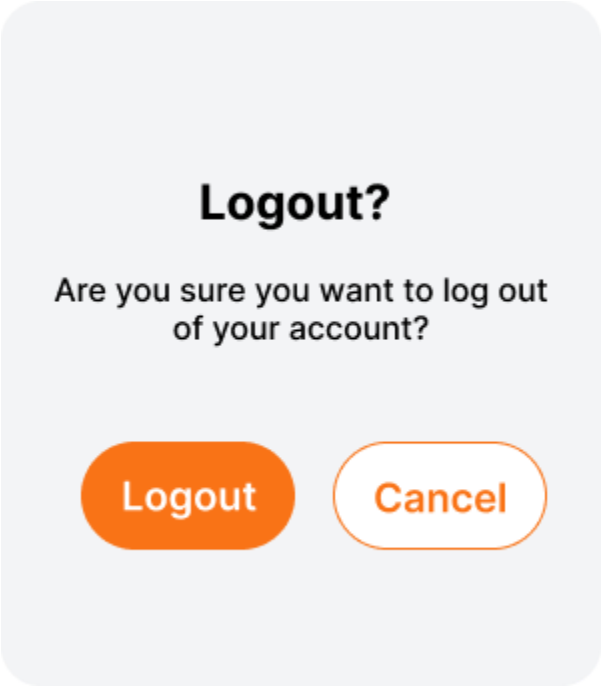
Logout Button:

- “Logout” Button (Placed below Account Info section)

Background:
Orange header section with light gray content area.

Footer:
Bottom Navigation Bar

Logout Modal



Modal Title:
"Logout?"

Message:
"Are you sure you want to log out of your account?"

Buttons:

- Logout (Primary – Orange)
- Cancel (Outlined)

Background:
Dimmed overlay behind modal.

8.0 PLAN

8.1 Project Timeline

Phase 1: Planning & Design (Week 1–2)

Week 1: Requirements & Architecture

- Project setup and initial documentation
- Completion of Functional Requirements (FRS) and Non-Functional Requirements (NFR)
- System architecture planning and overall structure design

Week 2: Detailed Design

- API endpoint specification
- MySQL database design and entity relationship modeling
- UI/UX wireframe preparation
- Implementation plan finalization

Phase 2: Backend Development (Week 3–4)

Week 3: Foundation

- Spring Boot project setup with required dependencies
- Database configuration and entity creation (User, Exercise, Workout, Session)
- Implementation of authentication using PasswordEncoder

- Development of exercise CRUD endpoints
- Development of predefined workout endpoints

Week 4: Core Features

- Implementation of custom workout creation logic
- Development of workout session tracking
- Implementation of user profile and workout history endpoint
- Input validation and error handling improvements
- API testing and refinement using Postman

Phase 3: Web Application Development (Week 5–6)

Week 5: Frontend Foundation

- React project setup
- Development of authentication pages (register and login)
- Implementation of exercise listing page
- Development of workout listing and detail pages
- Creation of custom workout builder interface

Week 6: Complete Web Features

- Implementation of workout session execution page
- Development of profile and workout history page

- Admin exercise management interface
- Responsive design adjustments
- API integration testing and debugging

Phase 4: Mobile Application Development (Week 7–8)

Week 7: Android Foundation

- Android project setup and structure configuration
- Development of authentication screens
- Implementation of exercise browsing screens
- Workout viewing and execution functionality
- API integration using Retrofit

Week 8: Complete Mobile App

- Custom workout creation feature
- Workout history and profile screen
- UI refinement and improvements
- Testing on emulator and physical device
- APK generation and documentation

Phase 5: Integration & Finalization (Week 9–10)

Week 9: Integration Testing

- End-to-end testing across web and mobile platforms
- Bug fixing and performance improvements
- Security review and validation checks
- Database optimization
- Documentation updates

Week 10: Finalization & Submission

- Backend final review and cleanup
- Web application final adjustments
- Mobile application final adjustments
- Final system testing
- Project submission preparation

Milestones

- M1: All design documents completed
- M2: Backend API fully functional
- M3: Web application integrated with backend
- M4: Mobile application integrated
- M5: Full system tested and finalized

Critical Path

- Authentication implementation
- Exercise and workout API completion

- Custom workout logic
- Frontend-backend integration
- Cross-platform testing

Risk Mitigation

- Start with the simplest working version of each feature
- Test integration points early and often
- Keep backup copies of stable versions
- Focus on core functionality before enhancements